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Ttimepiece having a driving wheel including a hole with projections disengageable with a crown shaft

Abstract

A timepiece includes an inner rotating ring including a plurality of teeth portions; a driving crown including a head portion and a shaft portion; and a driving wheel including a hole portion engageable with and disengageable from the shaft portion, and configured to mesh with the teeth portions. The hole portion includes a plurality of protruding portions disposed at intervals of an angle smaller than 90° and protruding toward a rotation axis of the shaft portion.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2021-112007, filed Jul. 6, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.
BACKGROUND

1. Technical Field
(2) The present disclosure relates to a timepiece.
2. Related Art
(3) For example, JP-A-2002-328183 discloses a wristwatch device including an inner rotating ring used as a display member disposed in a housing, an operation shaft that can be pushed into and pulled from the housing, and a clutch that transmits rotation of the operation shaft to the inner rotating ring.
(4) However, in the technique of JP-A-2002-328183, when a large torque is required to rotate the inner rotating ring, for example, when a heavy inner rotating ring, an inner rotating ring with an improved click feeling, or the like is used, an engagement portion between a driving crown corresponding to the clutch and a driving wheel may slide, and thus rotation of the driving crown may not be transmitted to the inner rotating ring. That is, even when the driving crown is rotated, the inner rotating ring may not be rotated.
SUMMARY
(5) A timepiece includes an inner rotating ring including a plurality of teeth portions; an operation part including a head portion and a shaft portion; and a driving wheel including a hole portion engageable with and disengageable from the shaft portion, and meshing with the teeth portions. The hole portion includes a plurality of projections disposed at intervals of an angle smaller than 90° and protruding toward a rotation axis of the shaft portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a plan view showing a configuration of a timepiece.
(2) FIG. **2** is a plan view showing an internal configuration of the timepiece.
(3) FIG. **3** is a plan view showing the internal configuration of the timepiece.
(4) FIG. **4** is a perspective view of the timepiece shown in FIG. **3** as viewed from a direction of approximate 2 o'clock.
(5) FIG. **5** is a cross-sectional view showing a configuration of the timepiece.
(6) FIG. **6** is a cross-sectional view taken along a line C-C′ of the timepiece shown in FIG. **3**.
(7) FIG. **7** is a cross-sectional view showing a configuration of an engagement portion between a shaft portion and a driving wheel shown in FIG. **6**.
(8) FIG. **8** is a perspective view showing a configuration of a first elastic member.
(9) FIG. **9** is a cross-sectional view taken along a line A-A′ of the timepiece shown in FIG. **3**.
(10) FIG. **10** is a perspective view showing a configuration of a second elastic member.
(11) FIG. **11** is a cross-sectional view taken along a line B-B′ of the timepiece shown in FIG. **3**.
(12) FIG. **12** is a cross-sectional view showing a configuration of an engagement portion according to a modification.
(13) FIG. **13** is a cross-sectional view showing a configuration of an engagement portion according to a modification.
DESCRIPTION OF EXEMPLARY EMBODIMENTS
(14) In each of the following drawings, three axes orthogonal to one another will be described as an X axis, a Y axis, and a Z axis. A direction along the X axis is referred to as an “X direction”, a direction along the Y axis is referred to as a “Y direction”, a direction along the Z axis is referred to as a “Z direction”, a direction of an arrow is referred to as a + direction, and a direction at an opposite side from the + direction is referred to as a − direction. The +Z direction may be referred to as “upper” or “upper side”, and the −Z direction may be referred to as “lower” or “lower side”, and a view from the +Z direction is also referred to as a plan view or a plane. Description is made on the assumption that a surface on the +Z direction is an upper surface and a surface on the −Z

direction at an opposite side from the +Z direction is a lower surface. In other words, it can be said that a direction from 3 o'clock to 9 o'clock of a timepiece is the X axis, a direction from 12 o'clock to 6 o'clock is the Y axis, and an axis orthogonal to the X axis and the Y axis is the Z axis.
(15) First, a configuration of a timepiece **100** will be described with reference to FIG. **1**.
(16) As shown in FIG. **1**, the timepiece **100** includes a flat and cylindrical case **10**. An inner rotating ring **20** and a dial **30** are disposed inside the case **10**. Hands **40** including a second hand, a minute hand, and an hour hand are disposed at the dial **30**. For example, a scale or the like is printed at a display surface side of the inner rotating ring **20**.
(17) A cover glass **50** is disposed on the case **10** to cover the inner rotating ring **20**, the dial **30**, and the hands **40**. Time display can be visually recognized from a front surface side of the timepiece **100** through the cover glass **50**. A surface on which the inner rotating ring **20** and the dial **30** are visually recognized is referred to as a display surface.
(18) Although not shown, a movement that drives the hands **40** is accommodated inside the case **10**. The movement includes a step motor and a wheel train that drive the hands **40**, and a control circuit board that controls the driving of the step motor. The movement may be a mechanical movement using a spring as a power source.
(19) Specifically, in a side surface of the case **10**, driving crowns **61** and **62** for adjusting and setting the movement, the hands **40**, the inner rotating ring **20**, and the like are disposed respectively in a 4 o'clock direction and a 2 o'clock direction.
(20) Next, configurations and functions of the inner rotating ring **20**, the driving crown **61** as an operation part, and elastic members **70** will be described with reference to FIGS. **2** to **5**.
(21) As shown in FIG. **2**, the timepiece **100** includes the driving crown **61** disposed in the 4 o'clock direction in the case **10**, a first elastic member **71** as one of the elastic members **70** disposed in a 12 o'clock direction, and a second elastic member **72** as the other of the elastic members **70** disposed in a 6 o'clock direction. That is, the second elastic member **72** is disposed at an opposite side in an in-plane direction of the inner rotating ring **20** from the first elastic member **71**. Materials of the elastic members **70** are, for example, a resin.
(22) The driving crown **61** includes a head portion **63**, a shaft portion **64** coupled to the head portion **63**, and a driving wheel **65** slidably coupled to the shaft portion **64**. The driving wheel **65** may not be included in a configuration of the driving crown **61**. When the head portion **63** is rotated in a state where the head portion **63** is pushed toward the case **10**, the driving wheel **65** does not rotate since the shaft portion **64** and the driving wheel **65** do not mesh with each other. On the other hand, when the head portion **63** is pulled in a direction away from the case **10**, the shaft portion **64** and the driving wheel **65** mesh with each other, and when the head portion **63** is rotated, the driving wheel **65** rotates. The shaft portion **64** is made of, for example, a metal material. The driving wheel **65** is made of, for example, a resin.
(23) FIG. **3** shows a state in which the circular-shaped inner rotating ring **20** is disposed on the case **10** shown in FIG. **2**. FIG. **4** is a perspective view of the timepiece **100** shown in FIG. **3** as viewed from a direction of approximate 2 o'clock. Specifically, FIG. **4** shows positional relationships among the inner rotating ring **20**, the first elastic member **71**, the second elastic member **72**, and the driving crown **61**.
(24) As shown in FIG. **4**, the inner rotating ring **20** is provided with a plurality of teeth portions **21** at a lower side (on a back surface at an opposite side from the display surface). As shown in FIGS. **3** and **4**, when the inner rotating ring **20** is disposed on the case **10**, the teeth portions **21** of the inner rotating ring **20** mesh with the driving wheel **65** of the driving crown **61**. The teeth portions **21** mesh with the first elastic member **71** and the second elastic member **72**.
(25) As shown in FIGS. **4** and **5**, in order to rotate the inner rotating ring **20**, the driving crown **61** as the operation part disposed in the 4 o'clock direction of the timepiece **100** is operated. When the driving crown **61** is pushed toward the case **10**, the driving wheel **65** and the shaft portion **64** do not mesh with each other, and thus the inner rotating ring **20** does not rotate even when the driving

crown **61** is rotated.
(26) On the other hand, when the driving crown **61** is pulled in a direction away from the case **10**, the driving wheel **65** and the shaft portion **64** mesh with each other. When the driving crown **61** is rotated, the head portion **63**, the shaft portion **64** penetrating the case **10** and coupled to the head portion **63**, and the driving wheel **65** coupled to the shaft portion **64**, which constitute the driving crown **61**, are rotated, and the inner rotating ring **20** is rotated.
(27) The elastic member **70** is used to prevent unintentional rotation of the inner rotating ring **20** caused by an impact or the like when the inner rotating ring **20** is not rotated, specifically, when the shaft portion **64** and the driving wheel **65** of the driving crown **61** are not meshed with each other (a state at a lower side in FIG. **5**).
(28) Specifically, as will be described later, a first protruding portion **75***a* of the first elastic member **71** (see FIG. **9**) meshes with the teeth portions **21** of the inner rotating ring **20** separately from the driving wheel **65**, so that unintentional rotation of the inner rotating ring **20** is prevented.
(29) As shown in FIG. **3**, the first elastic member **71** and the second elastic member **72** always mesh with the teeth portions **21** of the inner rotating ring **20**. Specifically, the driving wheel **65** is disposed in the 4 o'clock direction of the case **10**. As described above, the first elastic member **71** and the second elastic member **72** are respectively disposed in the 12 o'clock direction and the 6 o'clock direction of the case **10**. That is, the first elastic member **71** and the second elastic member **72** are disposed at positions that do not interfere with the driving wheel **65**.
(30) Next, a configuration of an engagement portion **80** where the shaft portion **64** and the driving wheel **65** of the driving crown **61** are engaged with each other will be described with reference to FIGS. **6** and **7**.
(31) FIG. **6** is a cross-sectional view taken along a line C-C′ of the timepiece **100** shown in FIG. **3**, in other words, a cross-sectional view as seen from the 4 o'clock direction. FIG. **7** is an enlarged view of the engagement portion **80**.
(32) As shown in FIG. **6**, as described above, the driving crown **61** includes the head portion **63**, the shaft portion **64** coupled to the head portion **63**, and the driving wheel **65** including a hole portion **65**A engageable with and disengageable from the shaft portion **64**. When the driving crown **61** is pushed toward the case **10**, the engagement portion **80** between the driving wheel **65** and the shaft portion **64** is disengaged from the shaft portion **64** and the driving wheel **65** (the state at the lower side in FIG. **5**). On the other hand, when the driving crown **61** is pulled in a direction away from the case **10**, the driving wheel **65** and the shaft portion **64** are engaged with each other at the engagement portion **80** (a state in FIG. **6** and a state at an upper side in FIG. **5**).
(33) When the head portion **63** is rotated in a state where the driving wheel **65** and the shaft portion **64** are engaged with each other, the teeth portions **21** of the inner rotating ring **20** move along with rotation of teeth portions **66** of the driving wheel **65**, and thus the inner rotating ring **20** rotates.
(34) As shown in FIG. **7**, the engagement portion **80** is a portion where the shaft portion **64** is engaged with the hole portion **65**A of the driving wheel **65**. The hole portion **65**A of the driving wheel **65** includes a plurality of protruding portions **65***a* as projections protruding toward a rotation axis **64**A of the shaft portion **64**, and a plurality of recessed portions **65***b* provided between adjacent protruding portions **65***a*. Specifically, the protruding portions **65***a* are disposed at intervals with an angle between the adjacent protruding portions **65***a* smaller than 90°. An angle θ between the adjacent protruding portions **65***a* is preferably, for example, equal to or larger than 40° and equal to or smaller than 80°. In the present embodiment, the protruding portions **65***a* are disposed at six positions, that is, at intervals of 60°.
(35) On the other hand, the shaft portion **64** is formed matching a shape of the hole portion **65**A. Specifically, the shaft portion **64** includes a plurality of recessed portions **64***a* recessed toward the rotation axis **64**A of the shaft portion **64**, and a plurality of protruding portions **64***b* provided between adjacent recessed portions **64***a*. Specifically, the recessed portions **64***a* are disposed at intervals with an angle between the adjacent recessed portions **64***a* smaller than 90°. An angle θ

between the adjacent recessed portions **64***a* is preferably, for example, equal to or larger than 40° and equal to or smaller than 80°. In the present embodiment, similarly to the protruding portions **65***a*, the recessed portions **64***a* are disposed at six positions, that is, at intervals of 60°.
(36) It is desirable that a gap between the protruding portion **65***a* of the hole portion **65**A and the recessed portion **64***a* of the shaft portion **64**, in other words, a gap between the shaft portion **64** and the hole portion **65**A is in a range in which mutual rattling is prevented as much as possible and the shaft portion **64** and the hole portion **65**A can mesh with each other.
(37) As described above, since the timepiece **100** includes the driving wheel **65** provided with the hole portion **65**A including the protruding portions **65***a*, and the driving crown including the shaft portion **64** engageable with and disengageable from the hole portion **65**A, a rotational force in a rotation direction of the shaft portion **64** can be reliably transmitted to the driving wheel **65** by using engagement between the protruding portions **65***a* and the recessed portions **64***a*. Therefore, when a large torque is required to rotate the inner rotating ring **20**, for example, even when a biasing force is applied to the inner rotating ring **20** by the elastic members **71** and **72**, sliding between the shaft portion **64** and the driving wheel **65** can be prevented. Therefore, the inner rotating ring **20** can be rotated along with the rotation of the driving crown **61**.
(38) The protruding portions **65***a* are disposed at intervals of a predetermined angle, and thus the shaft portion **64** and the hole portion **65**A can be engaged with each other even when a rotation amount of the driving crown **61** is small. In addition, since the shape of the protruding portions **65***a* is prevented from being extremely small or complicated, the strength of the protruding portions **65***a* can be maintained, and productivity of the protruding portions **65***a* can be prevented from being significantly reduced. When the angle θ exceeds 80°, an effect of preventing sliding between the shaft portion **64** and the driving wheel **65** tends to be weakened.
(39) In addition, an engagement amount between the hole portion **65**A including the protruding portions **65***a* and the shaft portion **64** including the recessed portions **64***a* can be doubly improved as compared with an engagement amount when the protruding portions **65***a* and the recessed portions **64***a* are not provided as in the related art, and thus the sliding between the shaft portion **64** and the driving wheel **65** can be prevented.
(40) Next, specific configurations and functions of the first elastic member **71** and the second elastic member **72** will be described with reference to FIGS. **8** to **11**.
(41) As shown in FIG. **8**, the first elastic member **71** includes a leg portion **73**, an extension portion extending from the leg portion **73**, and a protruding portion **75** coupled to the extension portion. Specifically, the leg portion **73** includes a first leg portion **73***a* and a second leg portion **73***b*. The extension portion is a beam portion **74** crossing over the first leg portion **73***a* and the second leg portion **73***b*. The protruding portion **75** is a first protruding portion **75***a* that is provided substantially at a center of the beam portion **74** and has a triangular shape in a side view.
(42) As described above, since the first protruding portion **75***a* is provided at the beam portion **74** crossing over the first leg portion **73***a* and the second leg portion **73***b*, the first elastic member **71** can be stabilized, and the first protruding portion **75***a* can mesh with the teeth portions **21** with a stable force. As a result, unintentional rotation of the inner rotating ring **20** can be prevented.
(43) FIG. **9** is a cross-sectional view along a line A-A′ of the timepiece **100** shown in FIG. **3**. As shown in FIG. **9**, the first elastic member **71** is disposed in a recessed portion **11** provided in the case **10**. Specifically, the recessed portion **11** includes a first recessed portion **11***a* that meshes with the first leg portion **73***a* of the first elastic member **71**, a second recessed portion **11***b* that meshes with the second leg portion **73***b* of the first elastic member **71**, and a third recessed portion **11***c* at which the beam portion **74** of the first elastic member **71** is disposed.
(44) The inner rotating ring **20** is disposed on the first elastic member **71**. Specifically, as described above, the inner rotating ring **20** is provided with the plurality of teeth portions **21** at a back surface side. The first protruding portion **75***a* of the first elastic member **71** meshes with one teeth portion **21** of the plurality of teeth portions **21** of the inner rotating ring **20**.

(45) The plurality of teeth portions **21**, for example, 60 teeth portions **21** are formed at a uniform pitch in a circumferential direction. That is, an angle between adjacent teeth portions **21** is 6° in a plan view. The number and the angle of the teeth portions **21** are not limited to this example. The first protruding portion **75***a* of the first elastic member **71** is formed in substantially the same shape as the pitch between the teeth portions **21**.
(46) In this manner, the teeth portion **21** of the inner rotating ring **20** meshes with the first protruding portion **75***a* of the first elastic member **71**, and thus unintentional rotation of the inner rotating ring **20** can be prevented even when the shaft portion **64** and the driving wheel **65** of the driving crown **61** do not mesh with each other. In addition, when the shaft portion **64** and the driving wheel **65** of the driving crown **61** mesh with each other, the teeth portions **21** and the first protruding portion **75***a* are in contact with each other at a regular interval when the inner rotating ring **20** is rotated in the circumferential direction, and thus a click feeling can be obtained.
(47) As shown in FIG. **10**, the second elastic member **72** includes a first leg portion **73***a*, a second leg portion **73***b*, a beam portion **74** crossing over the first leg portion **73***a* and the second leg portion **73***b*, and a second protruding portion **75***b* provided substantially at a center of the beam portion **74** and having a substantially trapezoidal shape in a side view. Specifically, the second protruding portion **75***b* has a flat portion longer than the pitch between the teeth portions **21** of the inner rotating ring **20**.
(48) FIG. **11** is a cross-sectional view taken along a line B-B′ of the timepiece **100** shown in FIG. **3**. As shown in FIG. **11**, the second elastic member **72** is disposed in a recessed portion **12** provided in the case **10**. Specifically, the recessed portion **12** includes a first recessed portion **12***a* that meshes with the first leg portion **73***a* of the second elastic member **72**, a second recessed portion **12***b* that meshes with the second leg portion **73***b* of the second elastic member **72**, and a third recessed portion **12***c* at which the beam portion **74** of the second elastic member **72** is disposed.
(49) The inner rotating ring **20** is disposed on the second elastic member **72**. Specifically, as described above, the inner rotating ring **20** is provided with the plurality of teeth portions **21** at a back surface side. The second protruding portion **75***b* of the second elastic member **72** is in contact with the plurality of teeth portions **21** of the inner rotating ring **20** in a manner of crossing thereover, and pushes the plurality of teeth portions **21** upward.
(50) In this manner, when the first protruding portion **75***a* of the first elastic member **71** meshes with the teeth portion **21**, the second protruding portion **75***b* of the second elastic member **72** pushes the teeth portion **21**, so that occurrence of rattling in the inner rotating ring **20** can be prevented. Further, since the inner rotating ring **20** is supported by the first elastic member **71** and the second elastic member **72**, balance of the inner rotating ring **20** can be maintained, and unintentional rotation of the inner rotating ring **20** can be prevented.
(51) As described above, the timepiece **100** according to the present embodiment includes the inner rotating ring **20** including the plurality of teeth portions **21**, the driving crown **61** including the head portion **63** and the shaft portion **64**, and the driving wheel **65** including the hole portion **65**A engageable with and disengageable from the shaft portion **64** and meshing with the teeth portions **21**, and the hole portion **65**A includes the plurality of protruding portions **65***a* disposed at intervals of an angle θ smaller than 90° and protruding toward the rotation axis **64**A of the shaft portion **64**.
(52) According to this configuration, since the timepiece **100** includes the driving wheel **65** provided with the hole portion **65**A including the protruding portions **65***a*, and the driving crown **61** including the shaft portion **64** engageable with and disengageable from the hole portion **65**A, the rotational force in the rotation direction of the shaft portion **64** can be reliably transmitted to the driving wheel **65** by using the engagement between the protruding portions **65***a* and the recessed portions **64***a*. Therefore, even when a large torque is required to rotate the inner rotating ring **20**, the sliding between the shaft portion **64** and the driving wheel **65** can be prevented, a rotational force of the head portion **63** can be transmitted to the inner rotating ring **20** via the driving wheel **65**, and the inner rotating ring **20** can be rotated. In addition, since the protruding portions **65***a* are

provided, the shaft portion **64** and the driving wheel **65** can be reliably engaged with each other even when the shaft portion **64** is rotated by a small amount, and operability can be improved.
(53) In the timepiece **100** according to the present embodiment, the angle θ may be equal to or larger than 40° and equal to or smaller than 80°. According to this configuration, since the protruding portions **65***a* are disposed at a predetermined angle θ in the above range, the shaft portion **64** and the hole portion **65**A can be engaged with each other even when the rotation amount of the driving crown **61** is small. In addition, since the shape of the protruding portions **65***a* is prevented from being reduced, the strength of the protruding portions **65***a* can be maintained, and the productivity of the protruding portions **65***a* can be prevented from being significantly reduced.
(54) In the timepiece **100** according to the present embodiment, the shaft portion **64** may be made of a metal material, and the driving wheel **65** may be made of plastic. According to the configuration, since the protruding portions **65***a* are provided, the rotational force of the shaft portion **64** made of the metal material can be reliably transmitted to the plastic driving wheel **65** even when the metal material and the plastic are engaged with each other.
(55) In addition, the timepiece **100** according to the present embodiment may include the first elastic member **71** including the first protruding portion **75***a* that meshes with the teeth portions **21**. According to the configuration, since the first elastic member **71** is provided, the teeth portion **21** and the first protruding portion **75***a* can be in contact with each other at a regular interval when the inner rotating ring is rotated, and a click feeling can be obtained. In addition, since the first elastic member **71** is disposed, even when a large torque is required to rotate the inner rotating ring **20**, a rotational force of the driving crown **61** can be transmitted to the inner rotating ring **20** due to the provided protruding portion **65***a*, and the inner rotating ring **20** can be rotated.
(56) In the timepiece **100** according to the present embodiment, the elastic member **70** may include the first elastic member **71**, and the second elastic member **72** disposed at a position different from and at an opposite side from the first elastic member **71** in an in-plane direction of the inner rotating ring **20**. According to the configuration, since the first elastic member **71** and the second elastic member **72** are provided, the inner rotating ring **20** can be prevented from being inclined toward one direction, and the occurrence of rattling can be prevented.
(57) Hereinafter, modifications of the above embodiment will be described.
(58) As described above, the shape of the engagement portion **80** is not limited to a substantially hexagonal shape in which the protruding portions **65***a* are disposed at intervals of 60° as shown in FIG. **7**, and may be any shape with easy meshing and generated torque, and may be shapes shown in FIGS. **12** and **13**. FIGS. **12** and **13** are cross-sectional views showing shapes of engagement portions **180** and **280** according to modifications.
(59) In the engagement portion **180** according to the modification shown in FIG. **12**, a hole portion **165**A is formed to have a substantially octagonal shape. Protruding portions **165***a* are formed at intervals of 45° in the hole portion **165**A. As described above, a shape of a shaft portion **164** is formed matching the shape of the hole portion **165**A including the protruding portions **165***a*.
(60) In the engagement portion **280** according to the modification shown in FIG. **13**, a hole portion **265**A is formed to have a substantially wavy shape. Protruding portions **265***a* are formed at intervals of 45° in the hole portion **265**A. As described above, a shape of a shaft portion **264** is formed matching the shape of the hole portion **265**A including the projections **265***a*.
(61) In addition, the hole portions **65**A, **165**A, or **265**A and the shaft portions **64**, **164**, or **264** may have a hexagonal shape, an octagonal shape, or a polygonal shape having more angles, and the number of the protruding portions **65***a*, **165***a*, or **265***a* is not particularly limited as well.
(62) The engagement portions **80**, **180**, or **280** is not limited to having a shape of protruding toward the rotation axis **64**A of the shaft portions **64**, **164**, or **264**, and may have a shape protruding toward an opposite direction from the rotation axis **64**A. Further, the protruding portions **65***a* of the driving wheel **65** are disposed at intervals of an angle θ smaller than 90° and protrude toward the rotation axis **64**A of the shaft portion **64**, but a configuration may be adopted in which the recessed portions

65*b* of the driving wheel **65** is disposed at intervals of an angle θ smaller than 90°, and the protruding portions **65***a* of the driving wheel **65** protruding toward the rotation axis **64**A may be provided between the recessed portions **65***b* of the driving wheel **65** that are disposed at intervals.

Claims

1. A timepiece comprising: an inner rotating ring including a plurality of first teeth; an operation part including a head and a shaft, the operation part being configured to move along a rotation axis of the shaft by push and pull operations by a user; and a driving wheel that is ring-shaped, the driving wheel having a plurality of second teeth in an outer surface and a plurality of projections in an inner surface to form a hole, the hole being engageable with and disengageable from the shaft, the plurality of second teeth being configured to mesh with the plurality of first teeth at a first engagement position, wherein the plurality of projections are disposed at intervals of an angle smaller than 90° and protrude toward the rotation axis of the shaft, the shaft of the operation part is configured to engage with the plurality of projections of the driving wheel at a second engagement position, and the first engagement position is shifted from the second engagement position along the rotation axis in a plan view.
2. The timepiece according to claim 1, wherein the angle is equal to or larger than 40° and equal to or smaller than 80°.
3. The timepiece according to claim 1, wherein the shaft includes a plurality of recessed portions configured to mesh with the plurality of projections.
4. The timepiece according to claim 1, wherein the shaft contains a metal material, and the driving wheel contains plastic.
5. The timepiece according to claim 1, further comprising: a case in which the inner rotating ring and the driving wheel are disposed, wherein when the operation part is pulled in a direction away from the case, the driving wheel and the shaft are engaged with each other, and when the operation part is pushed toward the case, the driving wheel and the shaft are disengaged from each other.
6. The timepiece according to claim 1, further comprising: an elastic member including a protruding portion configured to mesh with the plurality of first teeth.
7. The timepiece according to claim 6, wherein the elastic member includes a first elastic member and a second elastic member, and the second elastic member is disposed at a position different from that of the first elastic member in an in-plane direction of the inner rotating ring.